

Computability: Exercise 10

Due: June 15th, 2026

Question 1

- a. Prove that A_{TM} is **NP-Hard**.
- b. Prove that every non-trivial language L ($L \neq \emptyset, \Sigma^*$) is **P-Hard**, i.e., $K \leq_p L$ for every $K \in P$.

Question 2

We define $\text{coNP} = \{\overline{L} \mid L \in \text{NP}\}$. Show that if $\text{coNP} \cap \text{NP-Complete} \neq \emptyset$ then $\text{NP} = \text{coNP}$. (In fact, it is unknown whether $P = \text{NP}$ and whether $\text{NP} = \text{coNP}$).

Question 3

The decision problem 2-H is given as follows:

Input: An undirected graph $G = (V, E)$ with $|V| = n$, and an integer k ($0 \leq k \leq n$).

Question: Can the vertex set V be partitioned into two disjoint subsets V_1 and V_2 (where $V = V_1 \cup V_2$ and $|V_1| = k$) such that both induced subgraphs $G[V_1]$ and $G[V_2]$ contain a Hamiltonian cycle?

Show that 2-H is NP-Complete.